

Algebra 1
Unit 14
Lesson 1 Practice

Name _____
Date _____
Period _____

1. A survey of Junior and Senior class students asked participants if they will be ordering a yearbook. The results of that survey are shown.
- 466 Seniors are planning to purchase a yearbook
 - 197 Juniors are not planning to purchase a yearbook
 - 904 participants are planning to purchase a yearbook
 - There were 1,746 participants in the survey

Complete the joint frequency table.

Student Yearbook Purchase Plans

	Purchasing Yearbook	Not Purchasing Yearbook	Total
Juniors	438	197	635
Seniors	466	645	1,111
Total	904	842	1,746

2. How many Juniors are planning to purchase a yearbook?
438 juniors
3. How many total students are not planning to purchase a yearbook?
842 students
4. How many Seniors were surveyed?
1,111 seniors

5. A local theme park conducted a survey to find out their favorite snacks among three different key demographics. The data of this survey is shown in the table below.

Children Under 18		
Popcorn	Turkey Leg	Ice Cream
388	176	495

Ages 18 – 35		
Popcorn	Turkey Leg	Ice Cream
512	699	370

Ages 36 +		
Popcorn	Turkey Leg	Ice Cream
905	732	109

Construct a two-way frequency table that shows all of the survey data.

Favorite Snacks	Under 18	18 – 35	36 +	Total
Popcorn	388	512	905	1,805
Turkey Leg	176	699	732	1,607
Ice Cream	495	370	109	974
Total	1,059	1,581	1,746	4,386

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Unit 14
Lesson 2 Practice

Name _____
Date _____
Period _____

1. At Southside Middle School, 705 students were asked which of the school academies they would like to be in as their first choice. Eighty-two seventh graders said they would prefer the STEM academy. A total of 224 students said they would like to enroll in the Arts academy as their first choice, of which 108 are 7th graders. Of the 280 eighth graders, 75 of them would choose the STEM academy as their first choice. There are 192 sixth graders at Southside. Of the 220 Business academy choices, 71 are sixth graders.

Complete the two-way table summarizing the data.

Southside Middle School Academy Preference

	STEM	Arts	Business	Total
6 th	104	17	71	192
7 th	82	108	43	233
8 th	75	99	106	280
Total	261	224	220	705

2. A recent study was conducted among students to determine their most recent score based on the time, in hours, the participants spent studying. The following two-way table shows the resulting data.

Complete the table.

Test Score Survey

	0 – 50%	51 – 79%	80 – 100%	Total
0 – 2 hours	4	14	2	30
2 – 4 hours	2	12	8	22
4 + hours	0	1	23	24
Total	6	27	43	76

3. What is the smallest marginal frequency in the table?

6 students who scored 0 – 50%.

4. What is the largest joint frequency in the table?

23 students who scored 80 – 100% and studied for 4 + hours.

5. Describe what the value 43 means in the context of the data.

Out of 76 students in this study, 43 students scored 80 – 100% on their most recent test.

6. Describe what the value of 2 means in the context of the data.

Out of 76 students in this study, 2 students who scored 0 – 50% on their most recent test studied for 2 – 4 hours.

Algebra 1
Unit 14
Lesson 3 Practice

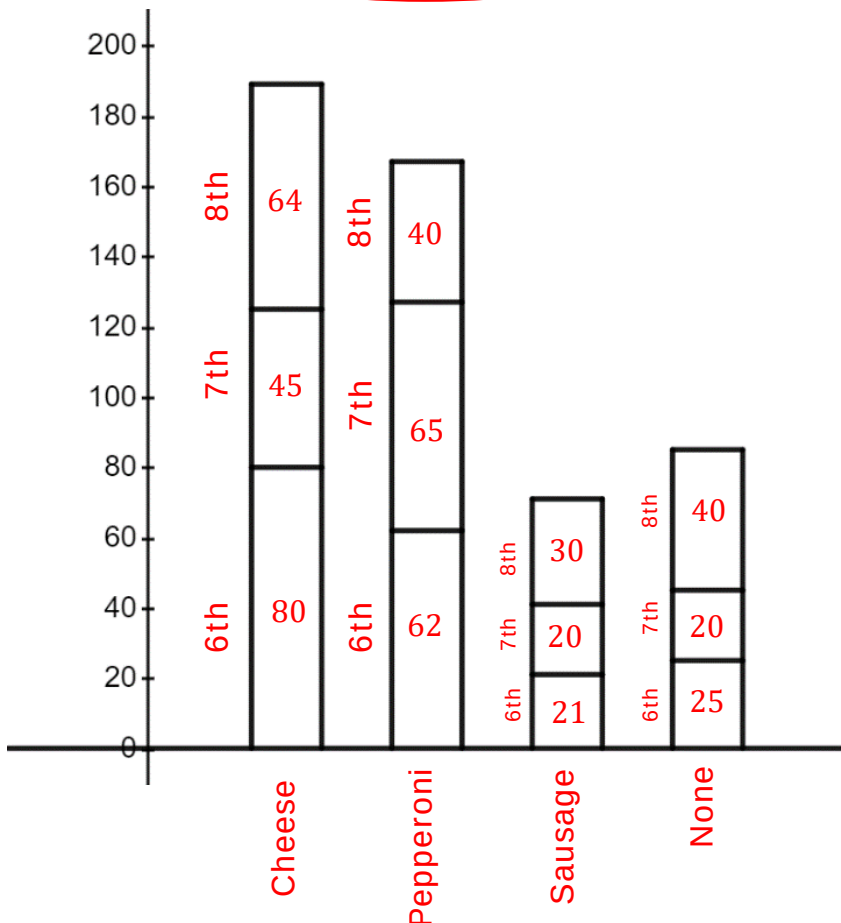
Name Answer Key
Date _____
Period _____

Southside Middle School conducted a school pizza party to celebrate the 100th day of school. Each student was given a choice of cheese, pepperoni, or sausage pizza. Some students did not want pizza. The results are shown in the frequency table below. A segmented bar graph was created with the data from the table. Use the figures to complete questions 1-5.

	6 th	7 th	8 th	Total
Cheese	80	45	64	189
Pepperoni	62	65	40	167
Sausage	21	20	30	71
None	25	20	40	85
Total	188	150	174	512

1. Circle the marginal frequencies in the table.

2. Box the joint frequencies in the table.



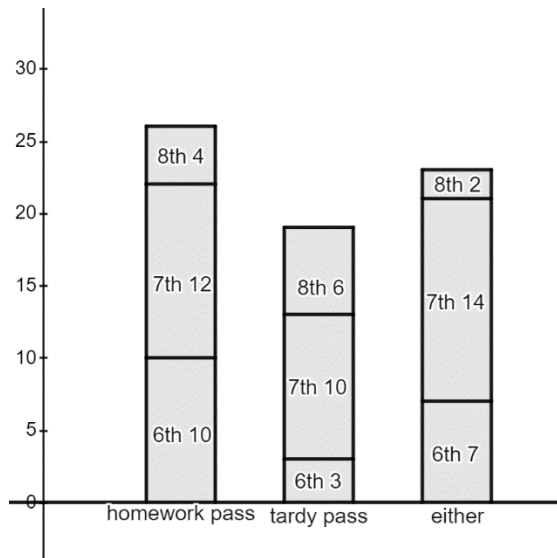
3. Label each marginal frequency in the boxes of the segmented bar graph.

4. Label the value for each joint frequency on the segmented bar graph.

5. Based on the data what type of pizza is the least favorite for each grade level?

Sausage

A teacher at a local middle school is deciding on an incentive for good behavior. The teacher polled students from each grade level to determine if students would prefer a tardy pass, homework pass, or either. The results are included on the segmented bar graph below. Use the bar graph to complete questions 6-10.



6. What does the joint frequency value of 12 represent on the bar graph?

The amount of 7th graders who prefer receiving a homework pass

7. How many 7th graders would prefer a tardy pass?

10

8. How many total students would prefer a homework pass?

26

9. How many 7th and 8th graders would like either reward?

16

10. If the teacher makes her decision based on the results in the segmented bar graph, what should she do? Explain your reasoning.

Homework pass because it is the most popular choice for 6th and 7th grade if you consider the votes for "either," it makes it a comparable choice for 8th grade as well.

Algebra 1
Unit 14
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

High school students in the United States were invited to complete an online survey in 2020. Part of the survey included questions about superhero powers. More than 1,000 students responded to this survey that included a question about a favorite superhero power. Researchers randomly selected 450 of the completed surveys. The two-way table broken down by gender gives partial results. Use this information to complete questions 1-6.

	To Fly	Freeze Time	Invisibility	Super Strength	Telepathy	Total
Females	49	60	48	1	70	228
Males	51	71	27	25	48	222
Total	100	131	75	26	118	450

- Complete the table.
- What fraction of male students chose Invisibility as their superpower?

$$\frac{27}{222} = \frac{9}{74}$$

- Does the fraction in #2 represent a marginal or joint frequency? Explain.

Joint (male and invisibility)

- What fraction of total students chose Super Strength as their superpower?

$$\frac{26}{450} = \frac{13}{225}$$

- Does the fraction in #4 represent a marginal or joint frequency? Explain.

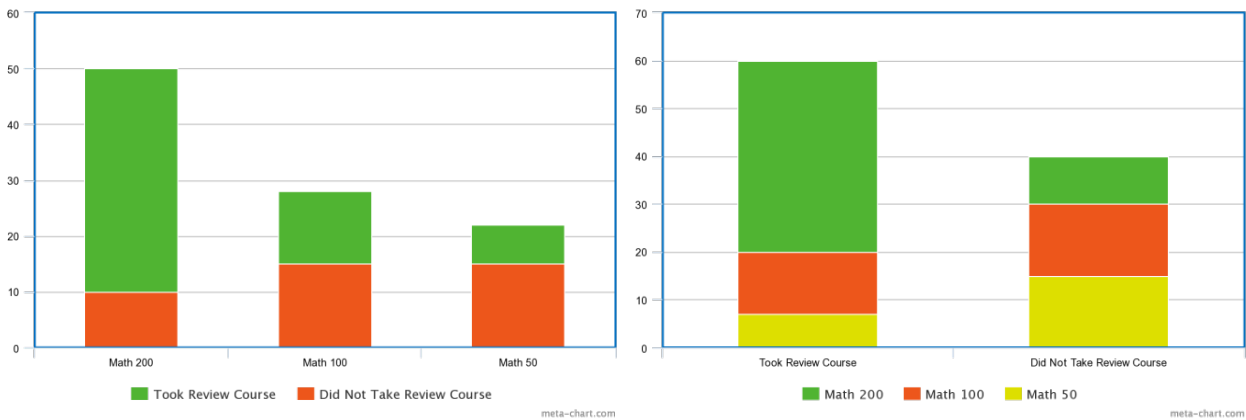
Marginal

- Is there evidence of an association between gender and a favorite superpower? Explain why or why not.

There are a few superpowers where there is a more significant difference between males and females choosing, so there is an association.

Students were given the opportunity to prepare for a college placement test in mathematics by taking a review course. Not all students took advantage of this opportunity. The following results were obtained from a random sample of students who took the placement test. The results are shown in the two way table and segmented bar graphs.

	Placed in Math 200	Placed in Math 100	Placed in Math 50	Total
Took Review Course	40	13	7	60
Did Not Take Review Course	10	15	15	40
Total	50	28	22	100



7. What fraction of the total surveyed were placed in Math 100?

 $\frac{28}{100} = \frac{7}{25}$

8. What fraction of students who took the review course were placed in Math 200?

 $\frac{40}{60} = \frac{2}{3}$

9. Is there evidence of an association between whether a student takes the review course and the math course in which the student was placed? Explain your answer.

Yes, because the majority of students in Math 200 took the review course.

10. Explain which data display was more helpful in determining evidence of an association between students who take the review course and the math course they are placed in.

The last segmented bar because you can more clearly see the increased number of students in Math 200 who took the review course.

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Unit 14
Lesson H1 Practice

Name Answer Key
Date _____
Period _____

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	To Fly	Freeze Time	Invisibility	Super Strength	Telepathy	Total
Females	49	60	48	1	70	228
Males	51	71	27	25	48	222
Total	100	131	75	26	118	450

1. Do you have enough information to create a two-way relative frequency chart from the information provided? Explain.

Yes, because you have the frequencies and totals for each category.

2. Complete the two-way relative frequency table to represent the data.

	To Fly	Freeze Time	Invisibility	Super Strength	Telepathy	Total
Females	10.9%	13.3%	10.7%	0.2%	15.6%	50.7%
Males	11.3%	15.8%	6%	5.6%	10.6%	49.3%
Total	22.2%	29.1%	16.7%	5.8%	26.2%	100%

Use the completed table from question 2 for questions 3–10.

3. What is the sum of the marginal relative frequencies?

100%

4. What is the sum of the joint relative frequencies for high school males?

49.3%

5. What is the sum of the joint relative frequencies for high school students who choose super strength as a superhero power?

5.8%

6. What percentage of those who chose telepathy are female?

15.6%

7. What percentage of females chose to freeze time?

13.3%

8. The values in question 6 and 7 represent

- ☐ joint
- ☐ marginal
- ☒ conditional

relative frequency.

9. Identify an example of a marginal relative frequency from the table.

Answers will vary.
Total high school students who chose “to fly”

10. Identify an example of a conditional relative frequency from the table.

Answers will vary.
Males who chose “to fly”

Algebra 1
Unit 14
Lesson H2 Practice

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Date _____
Period _____

High school students in the United States were invited to complete an online survey in 2020. Part of the survey included questions about superhero powers. More than 1,000 students responded to this survey that included a question about a favorite superhero power. Researchers randomly selected 450 of the completed surveys. The two-way table broken down by gender the results.

	To Fly	Freeze Time	Invisibility	Super Strength	Telepathy	Total
Females	10.9%	13.3%	10.7%	0.2%	15.6%	50.7%
Males	11.3%	15.8%	6%	5.6%	10.6%	49.3%
Total	22.2%	29.1%	16.7%	5.8%	26.2%	100%

1. What is the joint relative frequency for females who selected invisibility as their favorite superpower?

10.7%

2. What is the marginal relative frequency for freeze time?

29.1%

3. What is the difference in the joint relative frequencies for males and for females who selected to fly as their favorite superpower?

Males – Females

$$11.3 - 10.9 = 0.4$$

4. Interpret the value of 15.8% in the context of the data in the table.

15.8% of students are males who chose “Freeze Time” as their preferred superpower.

5. Interpret the value of 26.2% in the context of the data in the table.

26.2% of students chose telepathy as their preferred superpower.

6. Interpret the value of $\frac{11.3\%}{100\%}$ in context of the data in the table.

11.3% of the students surveyed were males who wanted to fly.

7. Interpret the value of $\frac{11.3\%}{49.3\%}$ in context of the data in the table.

22.9% of total males wanted to fly.

8. What percentage of females chose super strength as their favorite superpower?

$$\frac{0.2\%}{50.7\%} \approx 0.4\%$$

9. Is there a noticeable difference between the genders and their favorite superpowers?

Not a noticeable difference except with "Super Strength"

10. Is there evidence of an association between gender and a favorite superpower? Explain why or why not.

In general, because the relative frequencies don't all have a noticeable difference, there is not an association.